

#10 Abandoned Cart Recovery Email Automation

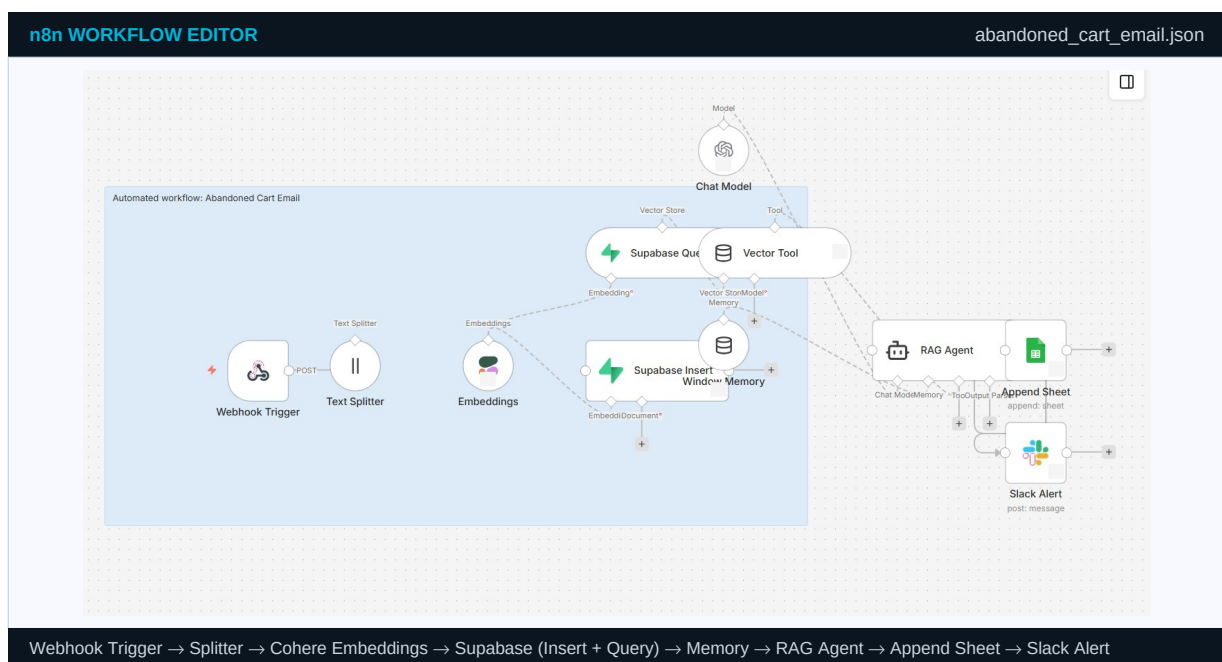
E-Commerce /
Revenue
Recovery

Recover lost revenue with hyper-personalised AI-written cart recovery emails

BUSINESS PROBLEM

70% of online carts are abandoned. Generic reminder emails recover only 3-5%. This workflow uses customer purchase history from a vector store to write truly personalised recovery emails mentioning specific items, referencing past purchases, and intelligently deciding whether to offer a discount.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Webhook POST — Shopify cart abandoned event (1h delay rule)

```
{
  "customer_id": "CUST-88271",
  "email": "sarah.k@email.com",
  "cart_items": [
    {
      "name": "Running Shoes X200",
      "sku": "RS-X200-BLK-7",
      "price": 89.99,
      "qty": 1
    },
    {
      "name": "Running Socks Pack",
      "sku": "SK-RUN-3PK",
      "price": 14.99,
      "qty": 2
    }
  ],
  "cart_value": 119.97,
  "abandoned_at": "2024-06-15T14:22:00Z",
  "session_id": "sess-abc123"
}
```

OUTPUT — Personalised email content + Sheets log entry

```
{
  "json": {
    "email_to": "sarah.k@email.com",
    "subject": "Sarah, your Running Shoes X200 are almost sold out",
    "discount_code": null,
    "discount_pct": 0,
    "abandon_count": 1,
    "email_version": "A",
    "sent_at": "2024-06-15T15:22:04Z",
    "personalisation": "referenced 2 previous Nike purchases",
    "predicted_conversion": 0.24
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Webhook — cart abandoned event	Shopify fires after 1h wait: {customer_id, email, cart_items[], cart_value, abandoned_at}. Returns 200 immediately.
2	⌘ Splitter — customer history chunks	Past orders, browsing sessions, and support interactions chunked for personalisation embedding.
3	■ Cohere Embeddings — customer profile	Behaviour patterns vectorised: 'bought running shoes 3x, prefers Nike, always buys mid-range, size 7'.
4	■ Supabase Insert — store cart context	Current cart stored as vector payload for agent reference during email generation.
5	■ Supabase Query — preferences fetch	'What has this customer bought before? What did they almost buy? Any brand or size preferences?'
6	■ Vector Tool — preference lookup	Fetches: discount history, return rate, preferred communication style, past ratings and reviews.
7	■ Window Memory — cart composition	Stores specific items so agent writes: 'the Running Shoes X200 in your cart' not 'items in your cart'.
8	■ RAG Agent — email generation and discount logic	Generates personalised email. Discount logic: 1st abandon = no discount, 2nd = 10% off, 3rd = 15% + free shipping.
9	■ Append Sheet — conversion tracking	Logs: sent_at, email_version, discount_applied, cart_value. Shopify pixel webhook updates with CONVERTED status.
10	■ Slack — high-value cart alert	Cart above \$500 fires Slack message to sales channel with customer details for personal phone outreach.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Cohere API Key	dashboard.cohere.com → API Keys	Cohere API
Supabase URL + Anon Key	app.supabase.com → Settings → API	Supabase API
OpenAI API Key	platform.openai.com/api-keys	OpenAI API
Slack Bot Token	api.slack.com/apps → OAuth & Permissions → Bot Token	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario	An online sports equipment store averages 340 cart abandonments per week with an average cart value of \$94. Their previous generic reminder email had a 4.2% recovery rate.
Input received	Sarah K. added Running Shoes X200 and Running Socks to cart, reached checkout, then left the site at 14:22. Shopify webhook fired at 15:22 after the 1-hour wait rule.
What the workflow found	Supabase showed Sarah had bought Nike running shoes twice in the past year. Cohere embeddings matched her profile: size 7, mid-range budget, prefers Nike. First abandonment — no discount warranted.

Output produced	Email sent at 15:22: 'Sarah, your Running Shoes X200 are almost sold out — only 3 pairs left in size 7. Based on your previous Nike purchases...' Sarah completed purchase at 16:44. Recovery rate for personalised emails in this store: 18.3% vs 4.2% generic.	
JSON: abandoned_cart_email.json	GitHub: trm-cart-recovery-agent	www.shopify.com/blog/abandoned-cart-email